

must change our spectrum policies. With a few exceptions, existing laws and regulations are rooted in historical anachronisms.

Since the passage of the Federal Radio Act in 1927 and the Communication Act in 1934, virtually everything about wireless has changed. What began as a technology for ship-to-shore communication became the foundation of the radio, broadcast television, satellite and cellular telephone industries, as well as supporting private radio services, public safety communications, military communications, wireless data networking and a host of other applications. The amount of spectrum considered usable has increased dramatically, as more sophisticated devices have been developed. Analog services are giving way to digital, allowing for additional features and efficiency.

Everything has changed except for one very important thing: We still regulate the radio spectrum based on the technology of the 1920s.

Spectrum licensing arose in the 1920s because of a historical phenomenon. Radio receivers of the period were primitive. They couldn't distinguish well between different transmissions, so the only way for multiple users to share the spectrum was to divide it up. In 1912, nearby ships hadn't responded to the Titanic's distress calls, prompting calls for regulation. By licensing spectrum to broadcasters, with wide separation between bands, the government could ensure that receivers could identify which signal was which. The exclusive licensing model was almost certainly the right approach when it was developed. It has been in place for so long, during which there has been so much commercial innovation in use of the wireless spectrum, that we take it for granted. When you think about it, though, our approach to spectrum is the exception rather than the rule. We shrug at intense government regulation of communications over the airwaves that would be unconstitutional in other media. After all, wireless communication is speech. Under the First Amendment, the government faces a high burden in justifying any law that